

Q1) one of the conditions is necessary for the signal  $x(t)$  to be periodic:

- A) the signal repeat itself at equal time interval
- B) the signal is composed of only one frequency
- C) the signal amplitude is always constant
- D) the signal is nonzero for all  $t$

Q2) for the signal  $x(t) = \sin(\frac{2\pi}{3}t)$  the fundamental period

- A) 3
- B)  $\frac{3}{2}$
- C)  $2\pi$
- D) 6

Q3) for  $x[n] = \cos(\frac{\pi}{3}n) + \sin(\frac{\pi}{4}n)$  is the signal periodic, and what is the fundamental period?

- A) yes, with a period of 12
- B) yes, with a period of 8
- C) not periodic
- D) yes, with a period of 24

Q5) what is the result of evaluating  $\int_{-\infty}^{\infty} e^{-t} \delta'(t) dt$ ?

- A) 0
- B) 1
- C) -1
- D) the integral is undefined

Q7) for  $\int_{-\infty}^{\infty} (t^2 + \cos(\pi t)) \delta(t-1) dt$  which value of

$t$  contributes to result?

- A)  $t = -1$
- B)  $t = 0$
- C)  $t = 1$
- D)  $t$  contributes over the interval

Q8) the inverse l.t of  $X(s) = \frac{1}{s+1}$ ,  $R_e(s) > -1$  is:  
 A)  $e^t$  B)  $e^{-t}$  C)  $t e^{-t}$  D)  $u(t)$

Q9) the inverse l.t of  $X(s) = \frac{s+1}{(s+1)^2 + 4}$ ,  $R_e(s) > -1$  is:  
 A)  $e^{-t} \sin(2t)$  B)  $e^{-t} \cos(2t)$  C)  $e^{-t}$   
 D)  $e^{-t} \cos(2t) + e^{-t} \sin(2t)$

Q10) the inverse l.t of  $X(s) = \frac{2s+4}{s^2+4s-3}$ ,  $R_e(s) > -1$  is:  
 A)  $2e^{-t} + 4e^{-3t}$  B)  $2e^{-t} - 4e^{-3t}$  C)  $4e^{-t} + 2e^{-3t}$   
 D)  $e^{-t} + e^{-3t}$

Q11) the  $x(t) = e^{-at} u(t) - e^{at} u(-t)$  is:  
 A) even B) odd C) neither D) periodic

Q12) ~~which~~ which of the following is true for F.S of  $x(t) = e^{-at} u(t) - e^{at} u(-t)$

- A) it contains only sine terms
- B) it contains only cosine terms
- C) Both sin and cos terms
- D) the FS Does not exist

Q13) the DC value for the Fourier series

of  $x(t) = e^{-at} u(t) - e^{at} u(-t)$  is:  
 A) 1 B) 0 C) 0.5 D) all are wrong

Q14) the inverse F.T of  $X(\omega) = \frac{1}{j\omega + 2}$  is:

- a)  $e^{-2t} u(t)$  ... B)  $e^{2t} u(t)$  ... c)  $e^{-2t} u(-t)$   
D)  $e^{2t} u(-t)$

Q15) what is the F.T  $X(f)$  of the  $x(t) = e^{-2t} u(t)$ ?

- A)  $\frac{1}{2 + 2j\pi f}$  ... B)  $\frac{1}{2 - 2j\pi f}$  ... c)  $\frac{2}{2 + 2j\pi f}$  ... D)  $\frac{1}{2 + jf}$